



(51) International Patent Classification:

C07D 279/12 (2006.01)

(21) International Application Number:

PCT/IN2021/050859

(22) International Filing Date:

06 September 2021 (06.09.2021)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

202021038419 05 September 2020 (05.09.2020) IN

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(81) Designated States (unless otherwise indicated, for every kind of national protection available): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, IT, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States (unless otherwise indicated, for every kind of regional protection available): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Declarations under Rule 4.17:

- as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))
- as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))

Published:

- with international search report (Art. 21(3))
- in black and white; the international application as filed contained color or greyscale and is available for download from PATENTSCOPE

(54) Title: PROCESS FOR PREPARING REVEFENACIN AND INTERMEDIATES THEREOF

(57) Abstract: The present invention relates to processes and intermediates for the synthesis of Revefenacin or pharmaceutically acceptable salts thereof.



WO 2022/049604 A1

**“PROCESS FOR PREPARING REVEFENACIN AND INTERMEDIATES  
THEREOF”**

**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to Indian Application No. **202021038419**, filed **5<sup>th</sup> September, 2020** and entitled “**PROCESS FOR PREPARING REVEFENACIN AND INTERMEDIATES THEROF**” which is incorporated herein in its entirety.

**FIELD OF INVENTION:**

The present invention relates to an improved process for the preparation of intermediates useful in the synthesis of Revefenacin or pharmaceutically acceptable salts thereof.

**BACKGROUND OF THE INVENTION:**

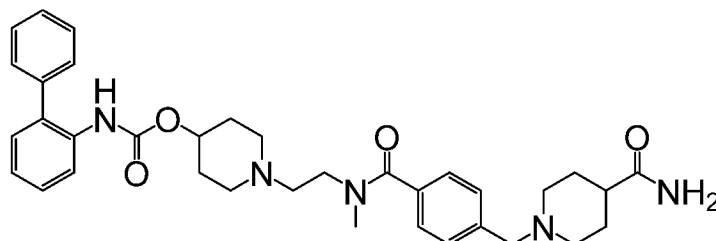
Pulmonary or respiratory disorders, such as chronic obstructive pulmonary disease (COPD) and asthma, afflict many millions of people worldwide and such disorders are a leading cause of morbidity and mortality.

Muscarinic receptor antagonists are known to provide bronchoprotective effects and therefore, such compounds are useful for treating respiratory disorders, such as COPD and asthma.

Revefenacin is a novel biphenyl carbamate tertiary amine agent that belongs to the family of the long-acting muscarinic antagonists (LAMA).

YUPELRI a sterile, clear, colorless, aqueous solution of Revefenacin, is an anticholinergic indicated for the maintenance treatment of patients with chronic obstructive pulmonary disease (COPD).

The chemical name for Revefenacin is 1-(2-{4[(4-carbamoylpiperidin-1-yl)methyl]-N-methylbenzamido}ethyl)piperidin-4-yl N-({1,1'-biphenyl}-2-yl)carbamate; its structural formula (I) is:

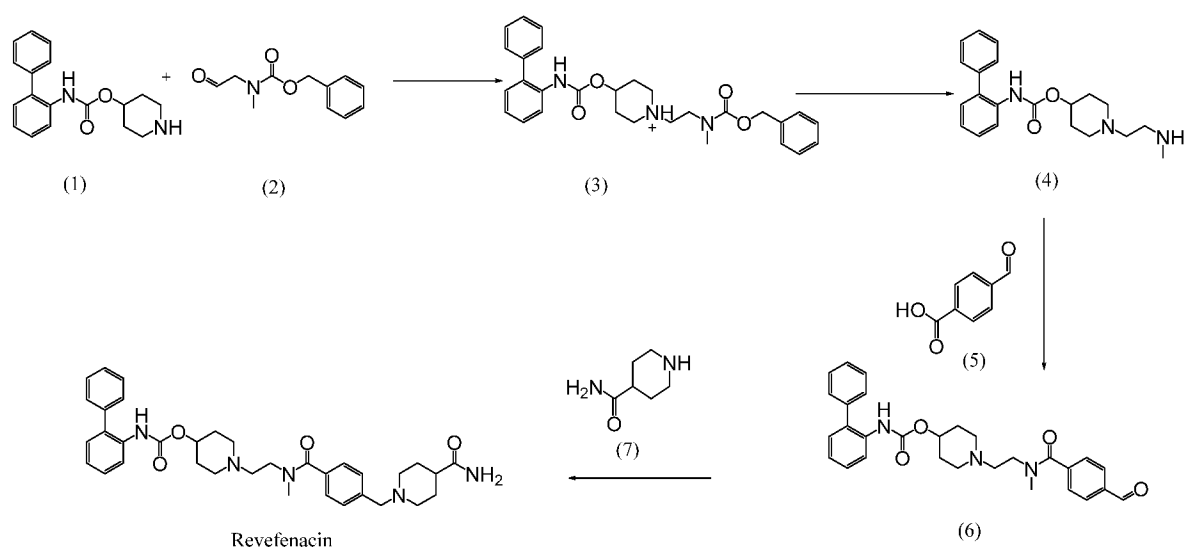


Revefenacin

US patent numbers US 7,288,657 B2 and US 7,585,879 B2 disclose novel biphenyl compounds having muscarinic receptor antagonist or anticholinergic activity to treat pulmonary disorders. In particular, Revefenacin and its process of preparation was first time specifically disclosed in these patents.

The process include oxidation of 2-(benzylmethylamino)ethanol to the aldehyde intermediate (2) followed by reductive amination with biphenyl-2-yl-carbamic acid piperidin-4-yl ester (1) and debenzylation, as depicted in reaction scheme 1 below:

**Scheme 1**



However, this process is not suitable on the industrial scale due to instable aldehyde intermediate (2) and low yield.

A refinement of the process is disclosed in the US 9,035,061 B2.

The process comprising the steps of:

- (a) reductive amination of a compound of formula (1): and a compound of formula (2) in the presence of a reducing agent, to yield the compound of formula (3);
- (b) debenzylation of the compound of formula (3) to yield the compound of formula (4);

wherein step (a) and step (b) are conducted in the same reaction mixture without isolation of the intermediate from step (a);

- (c) coupling a compound of formula (4) in the presence of a coupling reagent with a compound of formula (5) to yield the compound of formula (6);
- (d) reductive amination of the compound of formula (6) and a compound of formula (7) in the presence of a reducing agent to yield the compound of formula I;

wherein azeotropic distillation of water is conducted at an elevated temperature prior to the addition of the reducing agent, and reductive amination is conducted at room temperature; and

wherein step (c) and step (d) are conducted in the same reaction mixture without isolation of the intermediate reaction product from step (c).

Thus, there exists a continuous need for an efficient process of preparing Revefenacin and intermediates with high chemical purity and good overall yield, which is suitable for industrial scale up.

The present invention provides an improved process for synthesis of Revefenacin which avoids all the disadvantages associated with the prior art processes.

**OBJECTS OF THE INVENTION:**

One object of the present invention is to provide an improved process for preparing Revefenacin or pharmaceutically acceptable salts thereof.

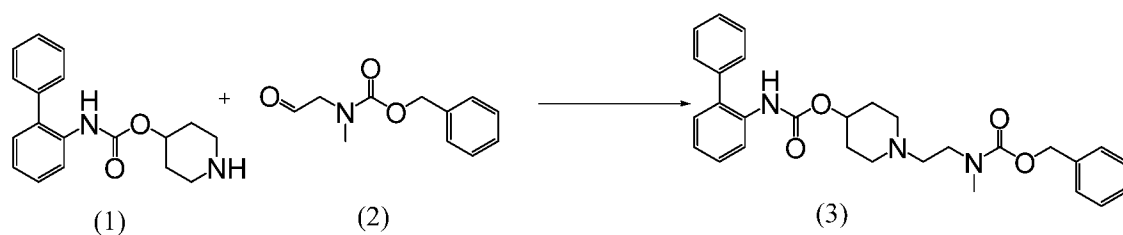
Another object of the present invention is to provide novel intermediates that are useful in the synthesis of Revefenacin or pharmaceutically acceptable salts thereof.

Another object of the present invention is to provide a purification method to obtain substantially pure Revefenacin or pharmaceutically acceptable salts thereof.

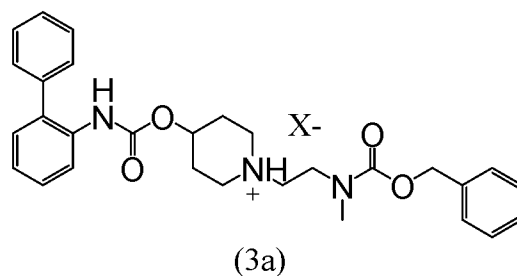
Yet another object of the present invention is to provide a process for preparing Revefenacin or pharmaceutically acceptable salts thereof which is simple, economical and suitable for industrial scale up.

**SUMMARY OF THE INVENTION :**

According to a first aspect of the present invention, there is provided a process for preparing compound of formula (3) or salt thereof comprising reductive amination of compound of formula (1) with compound of formula (2) in the presence of a reducing agent using suitable non polar solvent or mixtures thereof at 20-30°C and optionally converting compound of formula (3) to salt thereof.

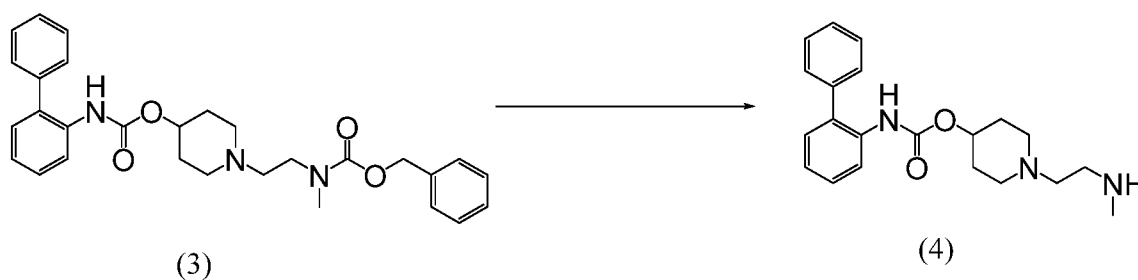


In an embodiment, the compound (3) is converted into an acid addition salt of the formula (3a) in the presence of an acid HX

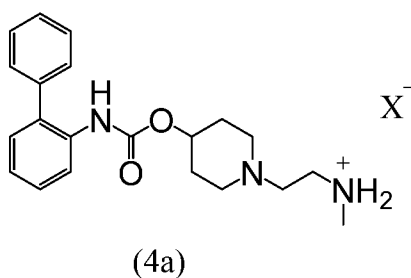


wherein  $X^-$  is an anion. The acid may be an inorganic acid or organic acid.

According to a second aspect of the present invention, there is provided a process for preparing compound of formula (4) or salt thereof comprising debenzoylation of compound of formula (3) or salt thereof by transfer hydrogenation under atmospheric pressure in the presence of a hydrogen donor and a hydrogenation catalyst and optionally converting compound of formula (4) to salt thereof.



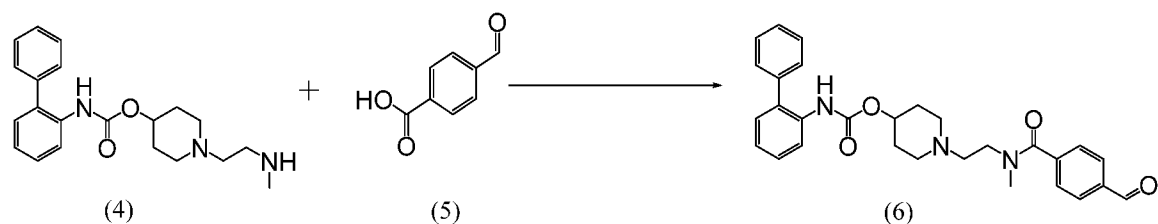
In an embodiment, the compound (4) is converted into an acid addition salt of the formula (4a) in the presence of an acid HX



wherein  $X^-$  is an anion. The acid may be an inorganic acid or organic acid.

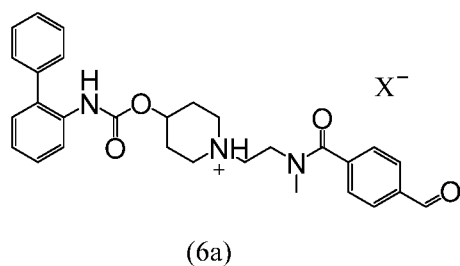
According to a third aspect of the present invention, there is provided a process for preparing compound of formula (6) or salt thereof comprising coupling of compound of formula (4) with compound of formula (5) in the presence of suitable

coupling reagent and a suitable inert ester solvent and optionally converting compound of formula (6) to salt thereof.



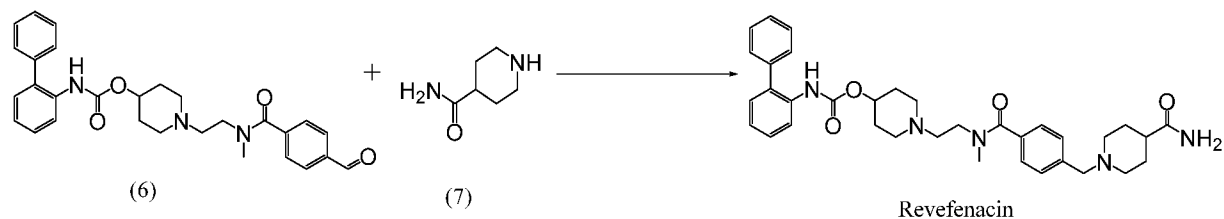
In an embodiment, the compound (6) is isolated as a solid.

In another embodiment, the compound (6) is converted into an acid addition salt of the formula (6a) in the presence of an acid HX

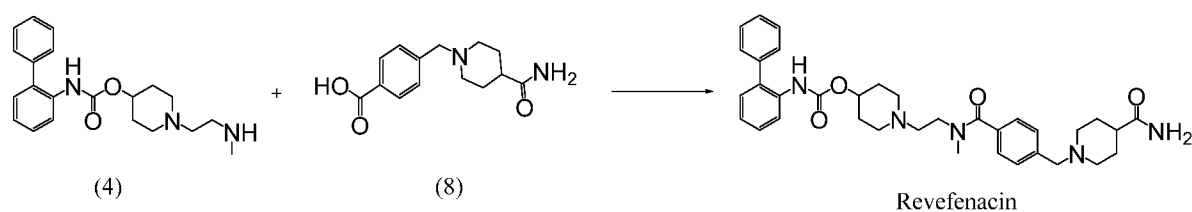


wherein  $X^-$  is an anion. The acid may be an inorganic acid or organic acid.

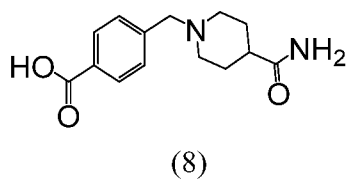
According to a fourth aspect of the present invention, there is provided a process for preparing Revefenacin of formula (I) or salt thereof comprising reductive amination of compound of formula (6) with compound of formula (7) in the presence of a suitable reducing agent and suitable aprotic solvent/s or mixtures thereof and optionally converting Revefenacin of formula (I) to salt thereof.



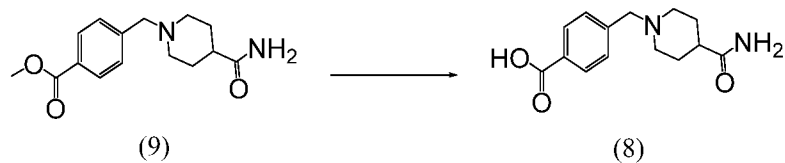
According to a fifth aspect of the present invention, there is provided a novel process for preparing Revefenacin of formula (I) or salt thereof comprising coupling of compound of formula (4) with compound of formula (8) in the presence of suitable coupling reagent and a suitable inert ester solvent and optionally converting Revefenacin of formula (I) to salt thereof.



According to a sixth aspect of the present invention, there is provided a novel intermediate compound of Formula (8)

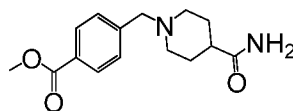


According to a seventh aspect of the present invention, there is provided a process for preparing compound of Formula (8) comprising hydrolyzing compound of formula (9) or salt thereof in the presence of a suitable base and suitable solvent/s or mixtures thereof.



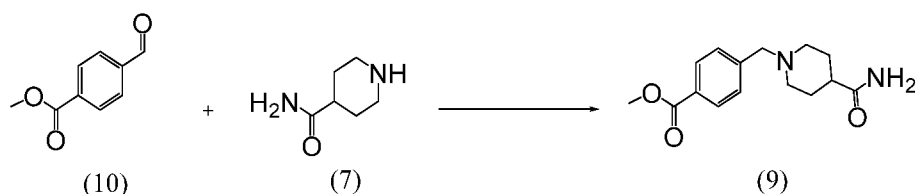
According to an eighth aspect of the present invention, there is provided a novel intermediate compound of Formula (9) or salt thereof





(9)

According to a ninth aspect of the present invention, there is provided a process for preparing compound of Formula (9) or salt thereof comprising reductive amination of compound of formula (10) with compound of formula (7) in the presence of a suitable reducing agent and suitable aprotic solvent/s or mixtures thereof.



According to a tenth aspect of the present invention, there is provided a process for stabilizing Revefenacin of formula (I) or salt thereof.

The process of the present invention has certain advantages over prior art

According to another aspect of the present invention, there is provided substantially pure Revefenacin of formula (I) or salt thereof having a purity of at least 95%, preferably at least 99%, more preferably at least 99.5%.

According to another aspect of the present invention, there is provided Revefenacin of formula (I) or salt thereof prepared according to a process defined above.

According to another aspect of the present invention, there is provided the use of Revefenacin of formula (I) or salt thereof prepared according to a process defined above in medicine.

According to another aspect of the present invention, there is provided the use of Revefenacin of formula (I) or salt thereof prepared according to a process defined above in treating pulmonary disorders.

The present invention also provides a method of treating a disease state prevented, ameliorated or eliminated by the administration of a long-acting muscarinic antagonists (LAMA) in a patient in need of such treatment, which method comprises administering to the patient a therapeutically effective amount of Revefenacin of formula (I) or salt thereof prepared according to the present invention, substantially as hereinbefore described.

Still another aspect of the present invention is to provide pharmaceutical composition containing a therapeutically effective amount of Revefenacin of formula (I) or salt thereof, prepared according to the present invention, along with one or more pharmaceutically acceptable carriers, diluents and excipients. Such pharmaceutical compositions are well known to those skilled in the art and processes for preparing them are also well known.

#### **Detailed Description of the Invention:**

The present invention provides an improved process for the synthesis of Revefenacin of formula (I) or salt thereof, as depicted in reaction scheme 2 below:

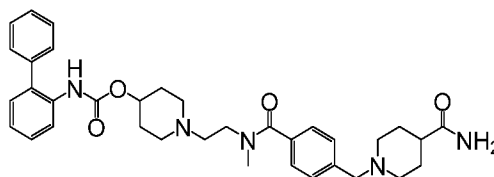
#### **Scheme 2**

water. The pH of the reaction was adjusted to 6.2 to 6.4 with 1% aqueous citric acid solution. Stirred at 20-25°C for 15 minutes and pH rechecked. The reaction mass further stirred at 20-25°C for 1 hour. The solids were isolated by filtration, washed with 2 volumes of sterile water and dried for 24 hours under vacuum at 40-45°C to yield Revefenacin (78-80g). HPLC purity > 99.8%

The product was packed in a virgin food grade double clear LDPE bags, tightly closed or heat sealed by purging nitrogen to prevent the contact with air and moisture. The packed polythene bag was further enclosed in a black polybag and TLHB bags, hermetically sealed and placed in a fiber drum. The pack was stored at 2 to 8°C.

**We Claims,**

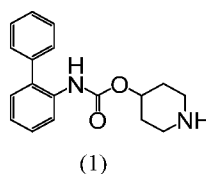
1. A process for the synthesis of Revefenacin of formula (I) or salt thereof,



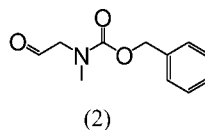
Revefenacin

the process comprising the steps of:

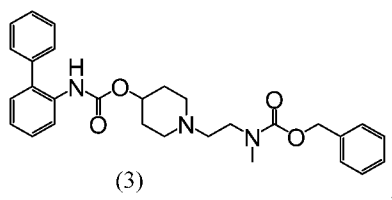
- (A) reductive amination of compound of formula (1)



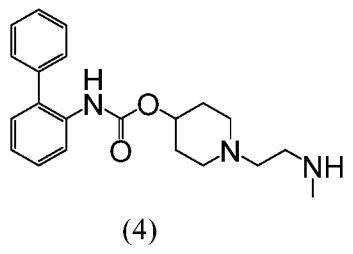
with compound of formula (2)



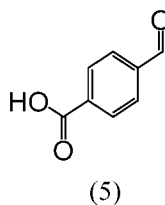
in the presence of a reducing agent and non polar solvent or solvent mixtures thereof to obtain compound of formula (3) or salt thereof



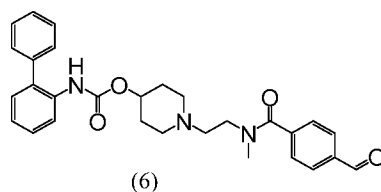
- (B) removing benzyloxy carbonyl group from compound of formula (3) under atmospheric pressure in the presence of a hydrogen donor and a hydrogenation catalyst, to obtain compound of formula (4) or salt thereof.



(C) coupling compound of formula (4) with compound of formula (5)

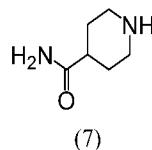


in the presence of a coupling reagent and an inert ester solvent to obtain amide compound of formula (6) or salt thereof



and ;

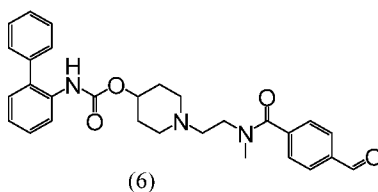
(D) reductive amination of compound of formula (6) with compound of formula (7)



in the presence of a reducing agent and aprotic solvent/s or mixtures thereof to obtain Revefenacin of formula (I) or salt thereof.

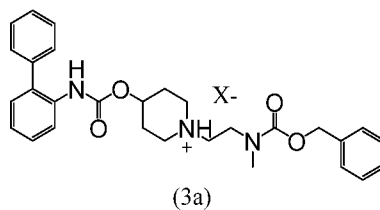
2. The process of claim 1, wherein the reducing agent in step (A) is selected from metal hydride reagents and borane reducing agents.
3. The process of claim 2, wherein the metal hydride reducing agent is sodium triacetoxo borohydride.
4. The process of claim 1, wherein non polar solvent in step (A) is selected from halogenated solvents and hydrocarbon solvents.
5. The process of claim 1, wherein step (A) is conducted at 20-30°C.

6. The process of claim 1, wherein the hydrogen donor in step (B) is selected from ammonium formate, formic acid, sodium formate, and potassium formate.
7. The process of claim 1, wherein the hydrogenation catalyst in step (B) is Group VIII metal catalyst.
8. The process of claim 1, wherein the coupling reagent in step (C) is 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride.
9. The process of claim 1, wherein the reducing agent in step (D) is selected from metal hydride reagents and borane reducing agents.
10. The process of claim 9, wherein the metal hydride reducing agent is sodium triacetoxy borohydride.
11. The process of claim 1, wherein steps (A), (B), (C) and (D) are conducted with or without isolation of intermediate compounds of formula (3), (4) and (6).
12. The process of claim 11, wherein compound of formula (6)

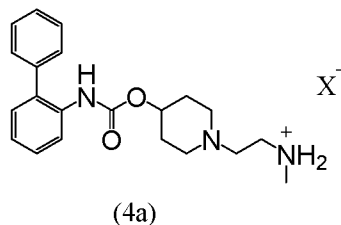


is isolated as a solid.

13. The process of claim 11, wherein compound (3) is isolated as an acid addition salt of the formula(3a)

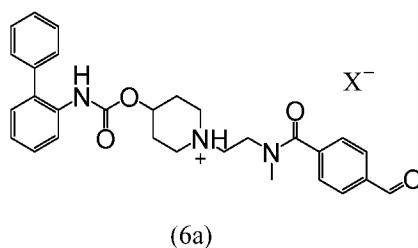


compound (4) is isolated as an acid addition salt of the formula (4a);



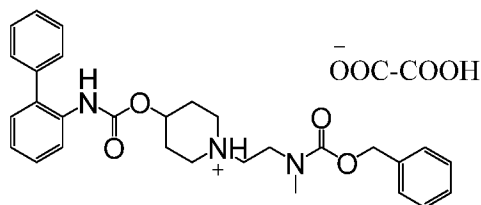
and;

compound (6) is isolated as an acid addition salt of the formula (6a)



wherein  $X^-$  is an anion;

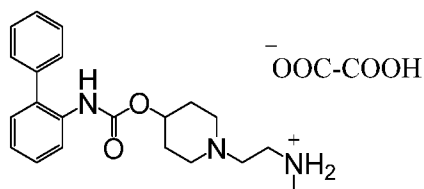
14. The acid addition salt according to claim 13, wherein anion corresponds to carboxylic acid selected from fumaric acid, tartaric acid, oxalic acid, succinic acid, tartaric acid, acetic acid maleic acid, malic acid, adipic acid, formic acid, malonic acid, phthalic acid, terephthalic acid, citric acid, methane sulfonic acid, ethane sulfonic acid, p-toluene sulfonic acid, trifluoroacetic acid or ascorbic acid and the like.
15. The acid addition salt of claim 13 or 14, wherein the acid is oxalic acid.
16. The acid addition salt of claim 15, wherein the compound (3) is compound of formula (3b)



(3b)

;

the compound (4) is compound of formula (4b)

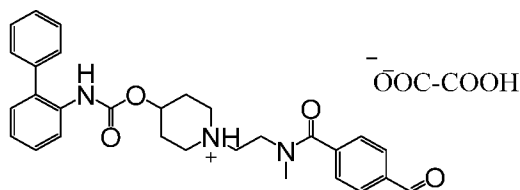


(4b)

;

and;

the compound (6) is compound of formula (6b)



(6b)

**17.** Substantially pure Revefenacin of formula (I) or salt thereof having a purity of greater than 99.8%.

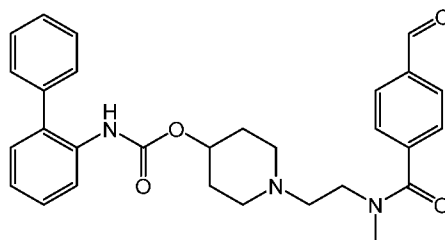
**18.** A process for the preparation of substantially pure Revefenacin of formula (I) or salt thereof comprising the steps of :

- a. Stirring Revefenacin of formula (I) in a suitable sterile solvent;
- b. Adjusting the pH to 6.2 to 6.4 with a buffer;
- c. Stirring for sufficient time;
- d. Isolating the solids; and
- e. Drying the solids.

**19.** The process of claim 18, wherein the sterile solvent is water.

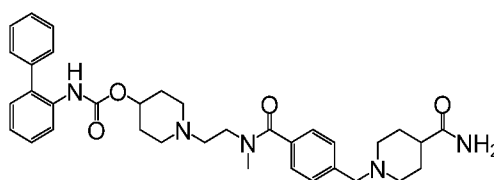


- 20.** The process of claim 18, wherein buffer comprises acid, selected from acetic acid, acetylsalicylic acid, adipic acid, alginic acid, ascorbic acid, aspartic acid, benzoic acid, benzenesulfonic acid, bisulfic acid, boric acid, butanoic acid, butyric acid, camphoric acid, camphorsulfonic acid, carbonic acid, citric acid, cyclopentanepropionic acid, digluconic acid, dodecylsulfic acid, ethanesulfonic acid, Ethylenediaminetetraacetic acid (EDTA), formic acid, fumaric acid, glyceric acid, glycine, gly-gly, glycerophosphoric acid, gluco heptanoic acid, gluconic acid, glutamic acid, glutaric acid, glycolic acid, hemisulfic acid, heptanoic acid, hexanoic acid, hippuric acid, hydrobromic acid, hydrochloric acid, hydroiodic acid, hydroxyethanesulfonic acid, lactic acid, maleic acid, malic acid, malonic acid, mandelic acid, methanesulfonic acid, mucic acid, naphthalenesulfonic acid, naphthilic acid, nicotinic acid, nitrous acid, oxalic acid, pelargonic, phosphoric acid, propionic acid, pyruvic acid, saccharin, salicylic acid, sorbic acid, succinic acid, sulfuric acid, tartaric acid, thiocyanic acid, thioglycolic acid, thiosulfuric acid, tosylic acid, undecylenic acid, MES, bis-tris methane, ADA, ACES, bis-tris propane, PIPES, MOPSO, cholamine chloride, MOPS, and combinations thereof, or salts thereof.
- 21.** The process of claim 20, wherein buffer comprises one or more of acetate, phosphate, sulfate, carbonate, formate, propionate, butanoate, lactate, glycine, maleate, pyruvate, citrate, aconitate, isocitrate,  $\alpha$ -ketoglutarate, succinate, fumarate, malate, oxaloacetate, aspartate, glutamate, tris(hydroxymethyl)aminomethane (tromethamine), combinations thereof, or salts thereof.
- 22.** The process according to claim 18 to 21, wherein the Revefenacin of formula (I) or salt thereof contains less than about 0.05% of the RFC 4-formyl biphenyl carbamate (Impurity A)



Impurity A

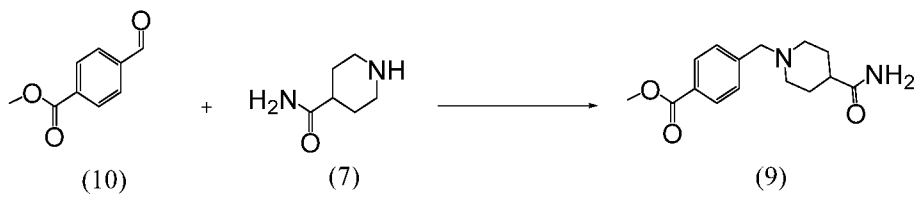
23. A process for the synthesis of Revefenacin of formula (I) or salt thereof,



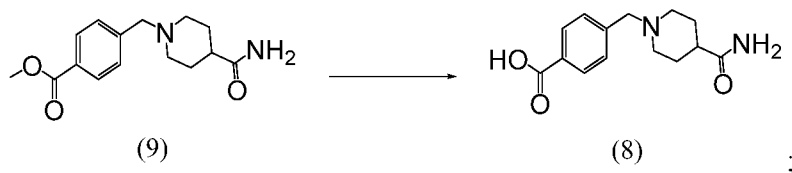
Revefenacin

the process comprising the steps of:

(E) reductive amination of compound of formula (10) with compound of formula (7) in the presence of a suitable reducing agent and suitable aprotic solvent/s or mixtures thereof to obtain compound of formula (9) or salt thereof



(F) hydrolyzing compound of formula (9) or salt thereof in the presence of a suitable base and suitable solvent/s or mixtures thereof to obtain compound of formula (8)



and;